

BHUMIKA PATIL

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EDUCATION

Degree / Exam	Institution	Score	Year
B.Tech – CSE	Sardar Vallabhbhai National Institute of Technology (NIT Surat)	8.70 CGPA	2023 – 2027
HSC (12th) – Science	Maharashtra State Board	91.5%	2023
SSC (10th)	Maharashtra State Board	92.0%	2021

EXPERIENCE

- **ReThink** – Software Development Intern May 2026 – July 2026 | Remote

Recycling Collection Centre Platform



React, TypeScript, Node.js/Express, MongoDB, Firebase

- Led product planning and system design – PRDs, data modeling, and RBAC – for a **5-role platform** (Admin, Aggregator, Centre, Recycler, Student) with independent dashboards and a phone-OTP onboarding flow backed by an admin approval queue for new centres, recyclers, and aggregators.
- Built the core **scan-to-reward engagement loop**: dual-mode image scanning for both logged-in users and unauthenticated QR-linked student scans with staff attribution, feeding a session-based cart that credits a points (*Karma*) ledger in real time.

SecondOpinion – Clinical Consultation Platform



NestJS, Next.js, PostgreSQL/Prisma, BullMQ, LiveKit

- Owned the core **case lifecycle** end-to-end – submission, specialist acceptance under SLA deadlines, mid-case **query/reply** threads, and **follow-up reopening** after closure – spanning a 14-status state machine with every transition logged to an immutable audit trail for compliance.
- Built multi-modal **case notes** supporting text, voice recordings, and image attachments via presigned uploads, plus **LiveKit video consultations** whose transcripts are auto-appended to the case and a no-show settlement engine that auto-refunds or credits either party.

PROJECTS

- **CollabCode: Real-Time Collaborative Code Editor**



React, Node.js, Express, Socket.IO, Yjs, BullMQ, Redis, Docker

- Engineered **real-time collaborative editing** for concurrent users across 18 languages using **Yjs CRDT**, ensuring conflict-free document synchronization and Google Docs-style cursor presence.
- Built a **Docker-based isolated code execution** pipeline spinning up language-specific containers per submission, executing untrusted code safely in sandboxed environments and auto-removing containers post-run.
- Designed a **fault-tolerant distributed architecture** using **BullMQ and Redis** as a persistent job queue across 3 independent worker nodes where crashed workers auto-retry jobs without data loss.
- Implemented **event-driven low-latency communication** via **Socket.IO** for real-time code updates, room chat, and user presence across multiple concurrent rooms.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C, C++, Python, SQL

Frontend: HTML, CSS, React.js, Next.js, EJS

Backend: Node.js, Express.js, NestJS, FastAPI, RESTful APIs, Socket.IO

Databases: MongoDB (Atlas), MySQL, PostgreSQL, Prisma

Version Control & Tools: Git, GitHub, Docker, Vercel, Render, Postman

Core Concepts: Computer Networks, OOP, DBMS, DSA, System Design